Michael Liu

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Education

University of Waterloo — *Sept 2025 - Apr 2030*
BCS in Computer Science, Digital Hardware Specialization

- Major Average: 93.1% | Cumulative Average: 92.7% | Student ID: 21177966
- Activities: Varsity Cross-Country, WAT.ai

Job Experience

Machine Learning Research Assistant — *Toronto, ON*
The Hospital for Sick Children (SickKids) — *May 2026 - Aug 2026*

- Conducted research on Reaction GFlowNets (RGFNs) for molecular generation and drug discovery.
- Designed and implemented experiments to evaluate molecular sampling strategies and generative model behavior.

Machine Learning Scientist — *Waterloo, ON*
WAT.ai — *Sept 2025 - Feb 2026*

- Collaborated with a small team of students on a medical imaging project focused on improving the segmentation of microaneurysms in fundus imaging.
- Co-authored a research paper detailing our methodology and results (GitHub).

Projects

C++ Machine Learning Library from Scratch — *github/cpp-ml-from-scratch*
C++, CMake

- Built a machine learning library from scratch in C++ centered around an n-dimensional tensor implementation, neural network layers, and classical machine learning algorithms such as Linear Regression, Logistic Regression, Softmax Regression, Decision Trees/Random Forests, Naive Bayes, and KNN.
- Prioritized transparency over abstraction by keeping the math behind forward passes, backpropagation, and parameter updates explicit through the training pipeline.

CourtVision: Computer Vision for Tennis Analytics — *github/tennis-cv-analysis*
OpenCV, PyTorch, YOLO, MediaPipe, XGBoost, Pandas, SciPy

- Built a computer vision system for tennis match analysis including player/ball tracking, court keypoint detection, and shot detection from broadcast camera footage.
- Implemented pose-based shot classification and computed player and ball speed statistics displayed through annotated match video output.

ArXtract: AI Search Engine for ML Research — *arxtract-cxc.vercel.app*
React, TypeScript, FastAPI

- Built a full-stack ML research tool using retrieval-augmented generation to query, rank, and summarize arXiv papers.
- Engineered semantic search using vector embeddings and cosine similarity to extract relevant sections from machine learning research papers based on user prompts.

Technical Skills

Programming Languages:	C/C++, Python, SQL, TypeScript
Data/Machine Learning:	PyTorch, scikit-learn, Pandas, NumPy, Matplotlib, OpenCV
Databases/Tools:	PostgreSQL, Git, GitHub, LaTeX, Excel, Tableau, CMake
Frontend:	React, HTML, CSS